

Creating a design for a Healthcare Information System that would allow Clayton State University Health Services to access student immunization records from Georgia Registry of Immunization Transactions and Services (GRITS) with consent of the student.

Tychira Brown

Introduction

The research aims to develop a design for a Healthcare Information System that enables Clayton State University Health Services to access student immunization records from Georgia Registry of Immunization Transactions and Services (GRITS), provided that the student gives their consent. It would allow for an improvement in the quality of care provided, and an enhancement in the accessibility of student services.

The design relates to Health Informatics, which focuses on managing and utilizing health information systems to enhance healthcare delivery and patient outcomes. Additionally, elements of the Public Health and Health Services Administration are pertinent, as they deal with access to care and health policy issues. I will utilize qualitative methods and draw on insights from various classes I have taken to design the Healthcare Information System.

It is important to improve the quality of care that students receive by facilitating access to their health information, which can lead to better health outcomes. It also seeks to streamline services at Clayton State University, potentially reducing waiting times and enhancing overall service delivery. By emphasizing the need for student consent, the study promotes transparency and builds trust in how health information is managed. Additionally, its findings could influence health policies and practices within the university and serve as a model for other institutions. Finally, by addressing the legal and ethical implications of sharing health information, the research ensures compliance with regulations while protecting student privacy. Overall, this study can potentially create a more effective healthcare system that supports students' academic and personal well-being.

It would develop the field of Health Informatics by offering a significant understanding of the designing and implementing of health information systems in educational settings. This research can open the door for further studies, encouraging other institutions to explore similar systems and adopting a collaborative spirit for improvement in student health services.

Globally, the study addresses the growth needed for efficient healthcare delivery in educational institutions, especially as student populations become more diverse. By focusing on the importance of accessible health information and informed consent, it highlights ethical practices

that can shape health policies beyond the university. This can lead to improved health outcomes and greater well-being for individuals everywhere.

The outcomes could directly impact student healthcare quality, influencing their health and academic success. It would address critical issues of privacy and consent, which are important for building trust between students and healthcare providers. The insights gained could benefit other educational institutions and healthcare systems, helping to improve student health services more generally. The research would emphasize systemic challenges in healthcare access, initiating important discussions about health equity and reform. This research would strengthen the understanding of the difficulties of health information management and its role in supporting student well-being.

Background and Significance

The research focuses on the lack of an efficient system to access student immunization records from Georgia Registry of Immunization Transactions and Services (GRITS) while prioritizing student consent. This gap negatively affects the quality of access to essential health services for students. The design of the Healthcare Information System would address this issue by enhancing access to vital health information for healthcare providers, ensuring that student consent and privacy are prioritized, and identifying best practices from existing systems.

The urgent need to improve student healthcare services at Clayton State University is affected by the demand for accessible and effective health services. It aims to bridge the gap between healthcare providers and essential student health information. The results could serve as a model for other institutions facing similar challenges, promoting the best practices in health informatics and student care more generally.

It would challenge the idea that improving access to student health information is just a logistical issue. Many current systems focus solely on integrating data sources, often neglecting the critical factors of student consent and privacy. By highlighting these aspects, the research will provide a more nuanced view of the problem, showing that simply having access to information is not enough for quality care. It will explore both the technical and ethical implications, considering student perspectives.

Aim and Research Questions

I aim to design a Healthcare Information System that enables secure and efficient sharing of student health information between Clayton State University Health Services and the Georgia Registry of Immunization Transactions and Services (GRITS), with a strong emphasis on student consent. This system will enhance access to care that will positively impacts their health and academic success. I will establish ethical guidelines for managing health information to foster trust between students and providers. The research will also involve analyzing existing healthcare information systems to identify the best practices and potential challenges, as well as evaluating the impact of the new system on student health outcomes. Ultimately, the goal is to contribute valuable insights to the field of health informatics, offering a model for other institutions and promoting best practices in student health services.

I want to conduct this research because I have seen firsthand how important access to quality healthcare is for students in a university setting. As a Resident Assistant and Graduate Resident Director, I observed how many students fought to balance their health with academic demands, often facing delays and complications when trying to access care, particularly for mental health issues. A memorable experience for me was during the COVID-19 pandemic when I worked in a dedicated COVID unit. Students were already dealing with stress and isolation and navigating the healthcare system added to their challenges. This experience made me realize how crucial efficient healthcare systems are in supporting student well-being. Through this research, I aim to contribute to a more effective healthcare system that helps students thrive academically and personally.

Research Design & Methods

I plan to utilize qualitative methods and draw insights from various classes I have taken to design the Healthcare Information System. I will review existing literature on healthcare information systems and student health services to identify gaps and inform my study design. I will conduct interviews with healthcare providers at student health services, and university administrators to gather perceptions of their experiences and needs. After analyzing the collected data and compiling it, I will then work on a design for the proposed Healthcare Information System.

One challenge I may face is navigating privacy concerns, particularly when dealing with sensitive health information. I will try to make sure that all research protocols meet the ethical guidelines and with emphasis on confidentiality to build trust with participants.

One limitation may be accessing existing data or information from existing healthcare systems. I will try to engage directly with administrators and stakeholders early in the research to identify any barriers and seek their support in facilitating access. By anticipating these challenges and having a plan in place, I aim to ensure a smooth research process.

Budget

To design the healthcare information system, I will have a \$0 budget. By focusing on only the essential features and using open-source tools for electronic health records. I will use university resources and collaborate with professors and peers for guidance and expertise. For security, will use free tools for encryption and rely on free resources to understand compliance basics. Testing can happen on local machines or through free cloud services, and I can gather feedback from healthcare students to help refine the system. I believe I can create a functional system using only available resources and free tools, keeping costs at zero.

Timetable

It will take 8 months with a \$0 budget to design the healthcare information system. I will break it down month by month.

- Months 1 and 2 would be to research the Georgia Registry of Immunization Transactions and Services (GRITS) and get insight into Clayton State University Health Services processes.
- Months 3 through 6 would focus on developing functional requirements, nonfunctional requirements, system proposal, database connection, interface diagram.
- Month 7 would focus on Refine requirements and diagrams.
- Month 8 would be aimed to finish up with PowerPoint, readme, and any other final documents.